

CITS4403-Project

due at 11:59 pm on Friday 9 October 2026

Checkpoint 1 — Weeks 7–8: Project idea and research question

Lab 2 | 241.203 COMPUTER SCIENCE.Computer Lab | 14 September 2:00–4:00 pm
| Facilitator: Warson

- Discuss your proposed system, research question, modelling approach, and overall project direction with your facilitator.

Proposed system

We propose to study bushfire spread through a simplified vegetation landscape, motivated by bushfire management in Australia, particularly Western Australia. We will focus on how local fire spread is influenced by the distribution of combustible vegetation and wind direction. The initial model will use a synthetic landscape rather than reproduce a specific real fire.

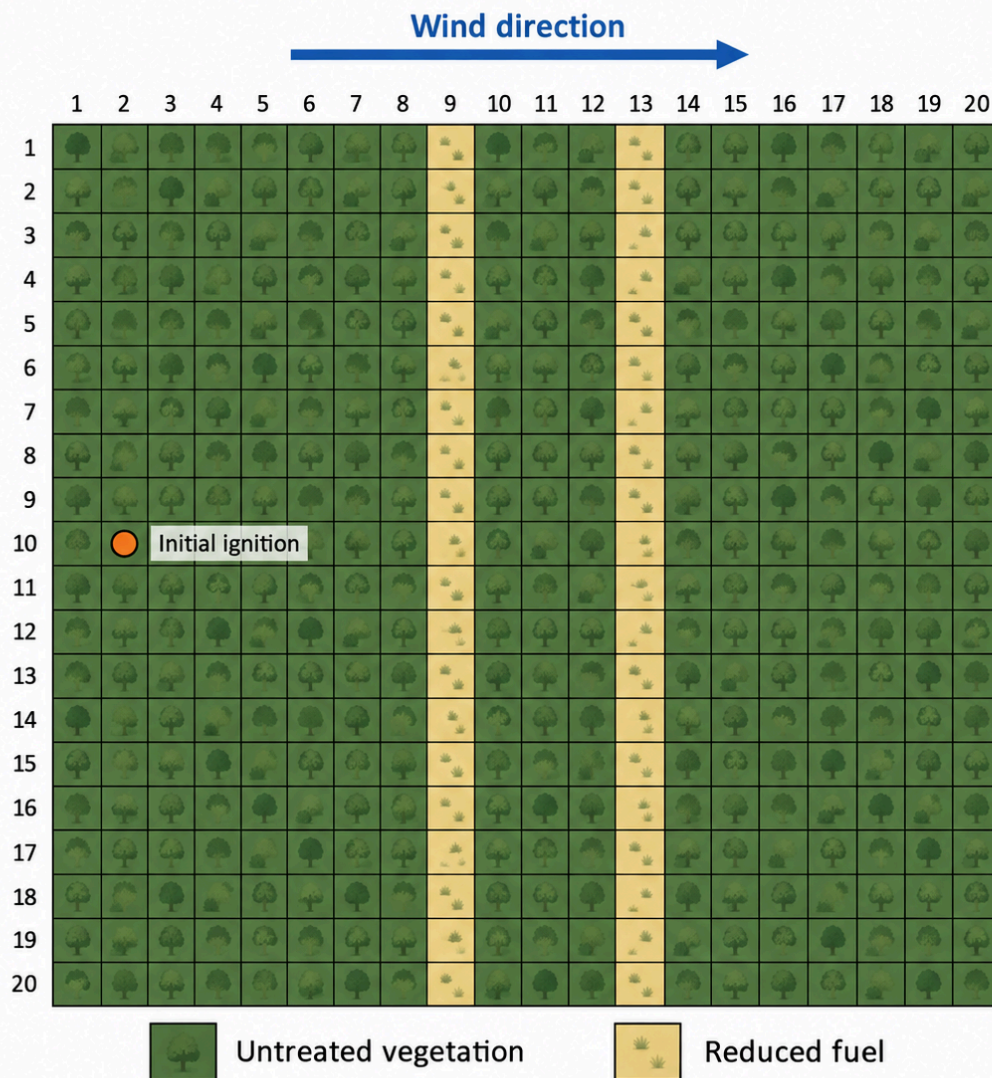
Research question

Our main research question is: Under an equal treated-area constraint, how do continuous fuel-reduction strips and dispersed treatment patches compare in reducing simulated bushfire spread, and how does their relative effectiveness change under different wind conditions?

Here, fuel treatment means reducing combustible vegetation. We will compare both layouts with an untreated baseline and examine whether their effects depend on the proportion of land treated.

A. Fuel-reduction strips

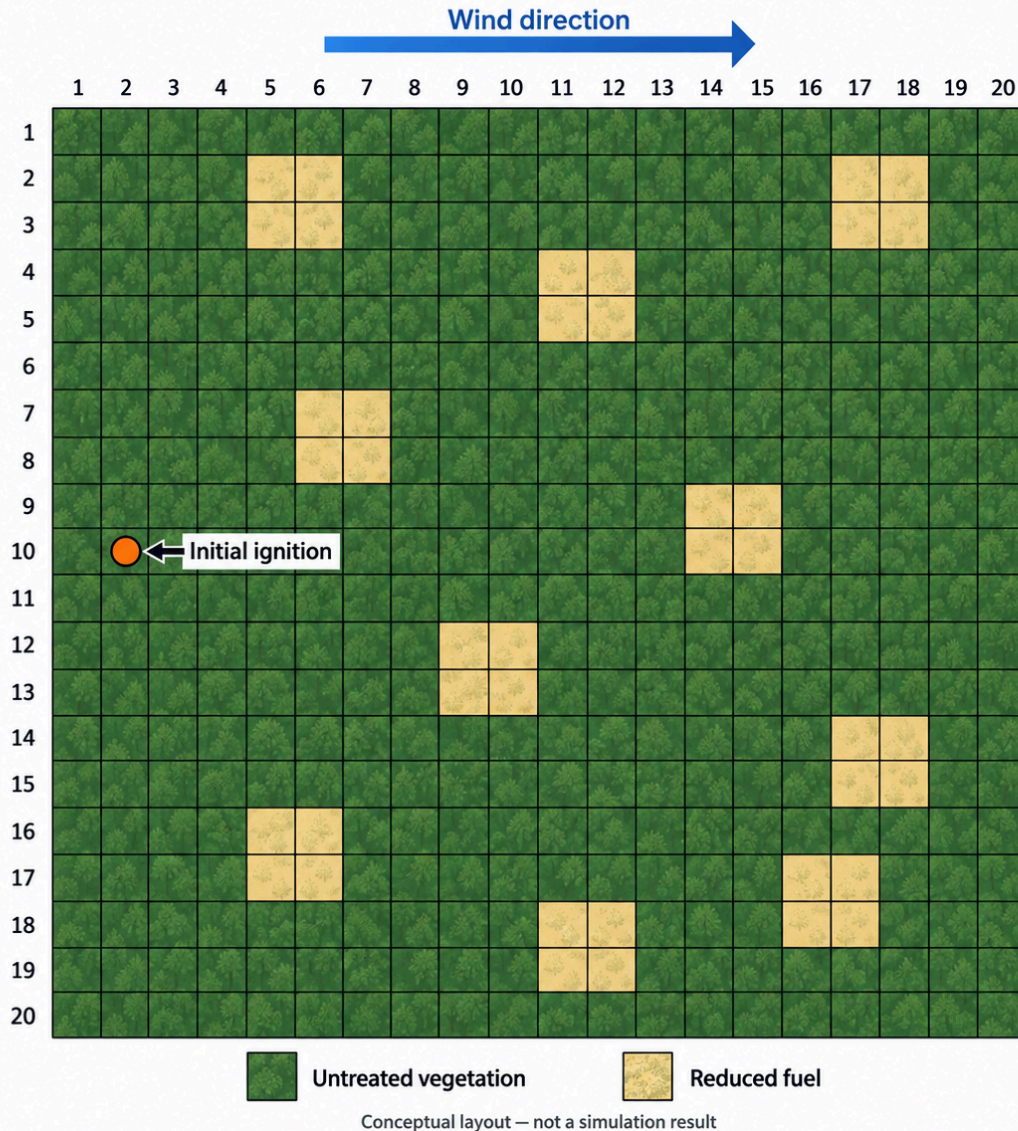
Continuous strips | 10% treated area



Conceptual layout — not a simulation result

B. Dispersed treatment patches

Dispersed patches | 10% treated area



Modelling approach

We plan to use a stochastic cellular automaton on a two-dimensional grid. Each cell will represent a patch of land with a state: unburnt, burning, or burnt. Fire will spread probabilistically between neighbouring cells, with ignition probability depending on the target cell's fuel state and the direction of spread relative to the wind.

Cells will update synchronously. We will initially assume a fixed wind condition within each simulation and a simple burning duration. We will use course material

and relevant literature to justify the spread rules and clearly distinguish modelling assumptions from empirically supported parameters.

Overall project direction

We will first implement and check a minimal fire-spread model, then introduce the two treatment layouts. We will keep treated area and treatment intensity equal when comparing layouts, and run repeated simulations across treatment proportions and wind conditions.

Our main outcomes will be the final burnt-area proportion and the probability of fire reaching the opposite side of the landscape. We will examine variability across runs and test whether the conclusions are sensitive to the assumed effectiveness of vegetation treatment.

The intended contribution is to investigate when one treatment layout performs better than another and whether that comparison remains robust under different conditions. We will discuss limitations, including simplified vegetation, fixed wind, and the absence of long-distance ignition. The report, reproducible code, and demonstration will present the same experiments and findings.

Feedback we would like

We would appreciate feedback on whether this scope provides sufficient independent investigation for the project, whether the modelling framework aligns with the unit, and whether a synthetic landscape is appropriate or some empirical calibration is expected.

/state: unburnt, burning, burnt

/cell: 50*50? 100*100?

/function: area sharp wind

/Change in combustion rate / Combustion area / Maximum combustion distance

- You should also demonstrate that your group has set up a shared GitHub repository for project collaboration.

Checkpoint 2 — Weeks 9–10: Model development and initial progress

- Demonstrate the main components of your model and discuss your experimental plan, progress, and any issues you have encountered.

Checkpoint 3 — Week 12: Project Demonstration

- Demonstrate your completed model, present your main findings, and answer questions about your implementation, experiments, and analysis.